Santiago Rodriguez

Portfolio

About Me

I'm a **mechanical engineer and product designer** focused on creating **efficient, intuitive systems** that improve how people interact with products. My approach starts with understanding the **core problem** and working backward to develop **clear, practical solutions** that meet **real needs**. I aim to make designs that are not just **functional, but also streamlined and effective** in real-world use.

I pride myself in my ability and resourcefulness in tinkering and rapid proof-of concept prototype making in addition to detailed scalable manufacturing focused designs

Project Overviews:

Project	Description	Note	Pages
GreenLoop	Invented a novel, automated hydroponic gardening appliance to define a new product category through rapid prototyping, manufacturing design and end-to-end system engineering.	Full Concept, Sketches, protoypes, and final product	3-18
Apple	Developed, designed, tested, and manufactured Next-Gen consumer device. Adding new experience defining features & developing novel cost-cutting architecture	Due to NDA, product is a conceptual adaptation No sketches or photos	19-31
P&G	Developing a novel product architecture to expand the Mr. Clean brand into a new category through industrial and mechanical design, user & performance testing, and manufacturing strategy	Due to NDA no sketches, photos, or specific product descriptions	32-38

GreenLoop

Bring Your Kitchen to *Life*

Effortlessly Fresh

Opportunity Space: Freshness vs Convenience

Context: The Modern Food System
Households rely on a retail supply chain designed for shipping durability rather than peak freshness. Produce is often harvested early and transported through long distribution networks before reaching the consumer.

Observed Consumer Behavior:
Studies show (next page) consumers want **fresh, high quality produce**, but their lifestyles prioritize **convenience and time efficiency**

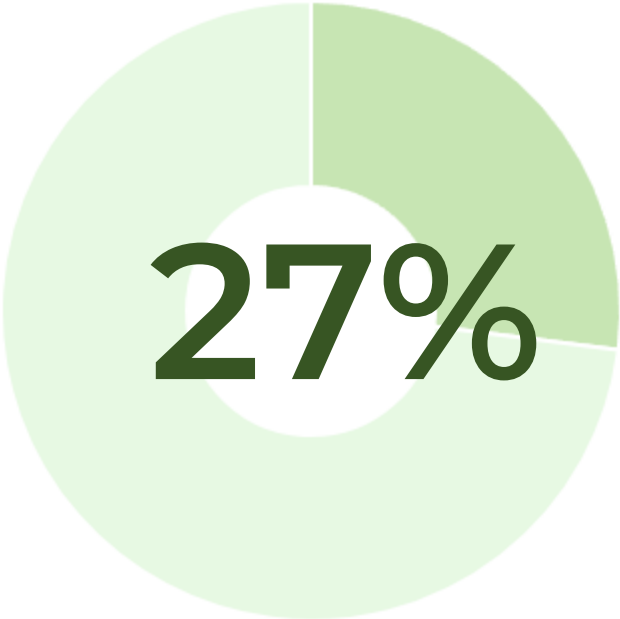
The current market treats freshness and convenience as opposing choices, creating a clear opportunity space

Option	Benefit	Limitation
Grocery retail	Convenient	Produce loses freshness in transit
Traditional gardening	Fresh ingredients	Requires time, space, and expertise

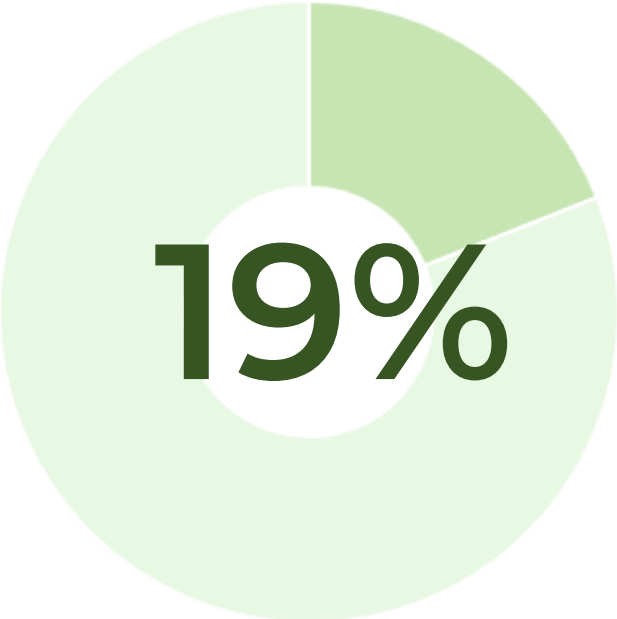
Observed Consumer Behavior

6/10

US Shoppers bought organic fresh produce at least **once** in the past year



Of US shoppers who buy organic once a year said they'd pay **10% more** for fresher produce



Of US shoppers who buy organic once a year said they'd pay **25%-49% more** for fresher produce

Root Cause Analysis:

Problem: Urban Household Rarely Grow/Buy Products to Grow Their Own Food

Why?

Home cultivation is high-risk and labor-intensive, often failing to deliver a reliable supply

Why?

Most households lack the time and expertise to manage feeding, watering, and lighting schedules.

Why?

The burden of keeping ingredients alive falls entirely on the user, not the home environment.

Why?

Current kitchen technology only slows spoilage after produce is harvested and shipped.

Why?

Retail supply chains prioritize durability over peak flavor, harvesting before ingredients are fully ripe.

Why?

Root Cause

Freshness depends on time and location, but homes lack a system that keeps produce at peak, ready for immediate use.

Greenloop is a smart hydroponic gardening kitchen appliance that provides effortless access to fresh ready-to-use vegetables, with a modular design that adapts to your needs.

GreenLoop:

Seamless Integration:
An ergonomic, intuitive design that fits naturally into a modern home ecosystem, becoming an essential every-day appliance

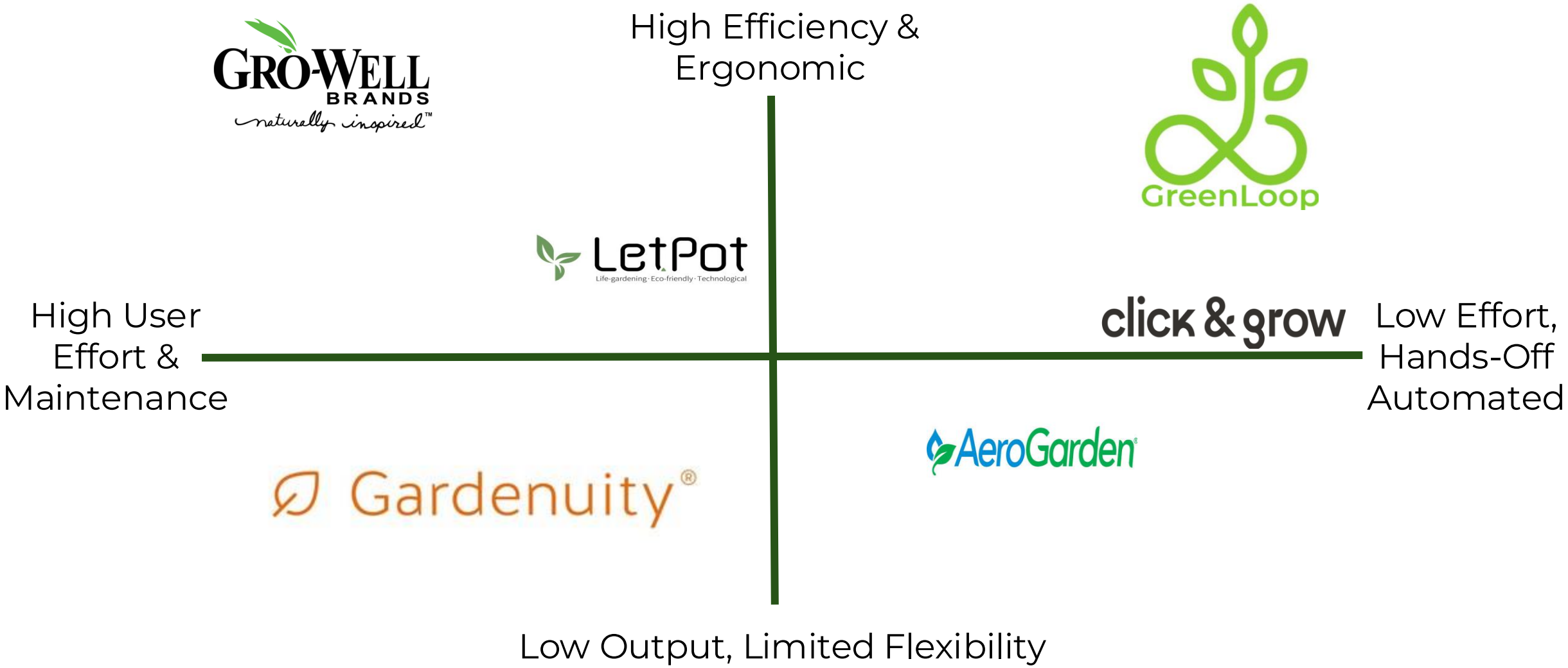
Modular & Scalable Design:
Start with a single unit and expand as your needs grow with GreenLoop's self-stacking & integrating design

High Variety & High Yield:
Innovative Hydroponic soil-free cycling system enables rapid plant growth and supports many plant types

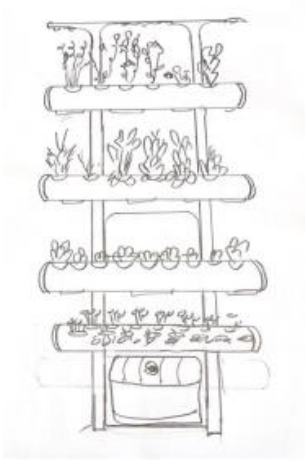
Automated Care:
Handles water circulation, feeding, and optimized lighting with minimal user intervention

Competitive Landscape:

Greenloop's efficient nutrient cycling system and ergonomic design stands out in the market



Prototype Development



Form Factor Exploration:

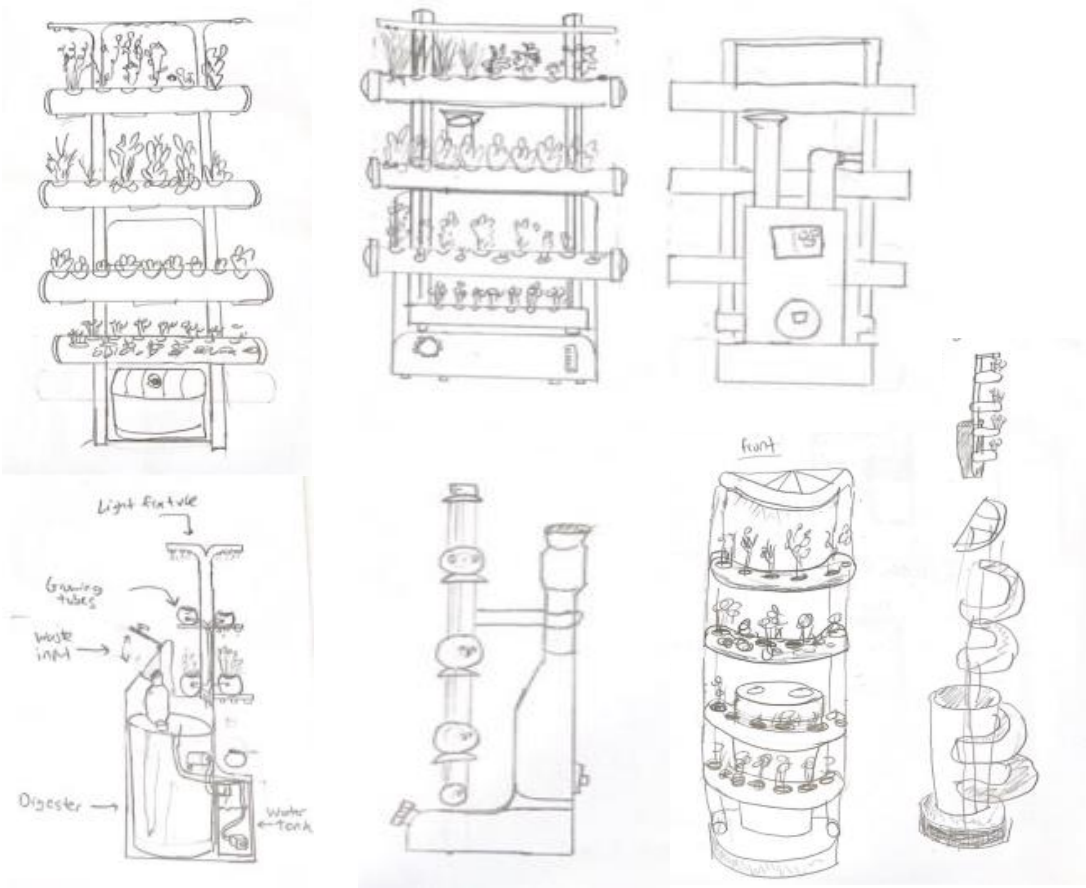
Defining the Ideal Form and Dimensions For Consumers

Approach:

- Explored multiple form factors for in-home integration
- Evaluated footprint, accessibility, visual presence and aesthetic design
- Considered common kitchen and apartment constraints

Industrial Design Considerations:

- Studied proportions and scale relative to existing kitchen appliances
- Explored unique design approaches, geometries, and profiles
- Explored geometry for both standalone presences and corner integration



The Challenge was not just functionality, but creating a design that balances presence, usability within a home without adding friction

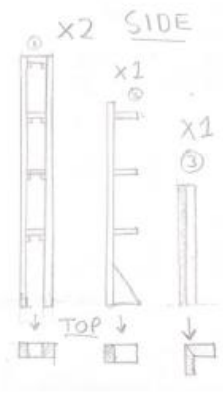
Building Initial Large-Scale Prototype

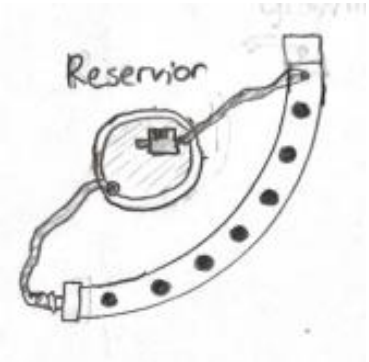
Designed (CAD) & Built (Rapid Prototyping) a Full Scale Prototype

Approach:

- **Built** a large (~4ft tall) corner fitting quarter-circle prototype resulting from my ideation
  - Utilized 3D printing, carpentry, machining, and a self-developed PVC heat-forming technique to rapidly produce cost-effective prototypes

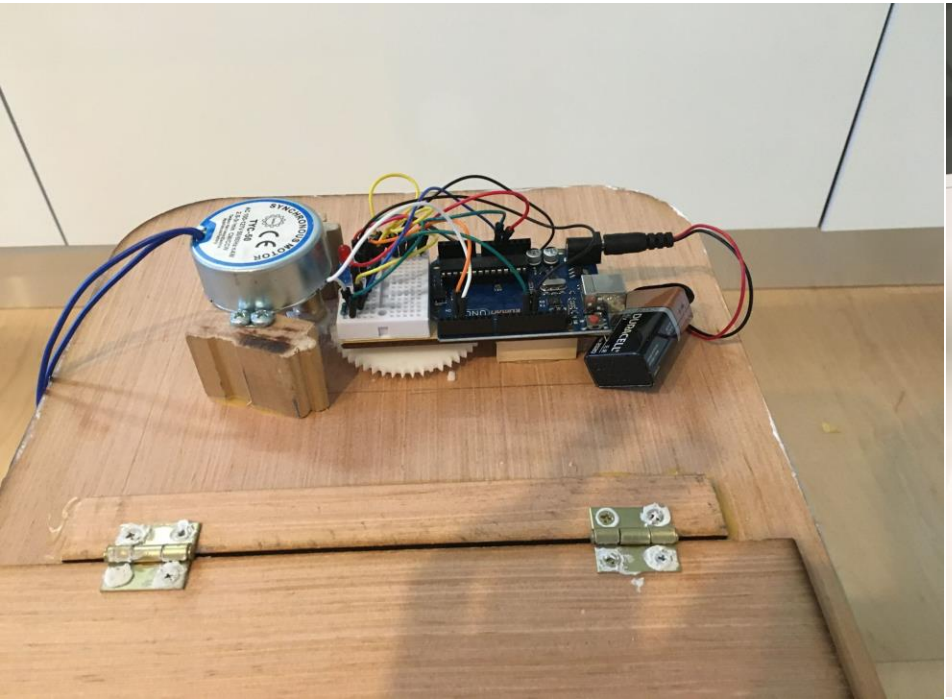
- Created my own PVC forming fixture out of wood and dowel pins to **quickly make prototype** components (within 24 hours)
- Worked in prototyping shop to quickly make multiple units for testing using **hands-on prototyping approach to speed up building process**
- Created **fully functional closed loop hydroponic flow system**, taking into account flow rates, and incorporating baffles to improve aeration

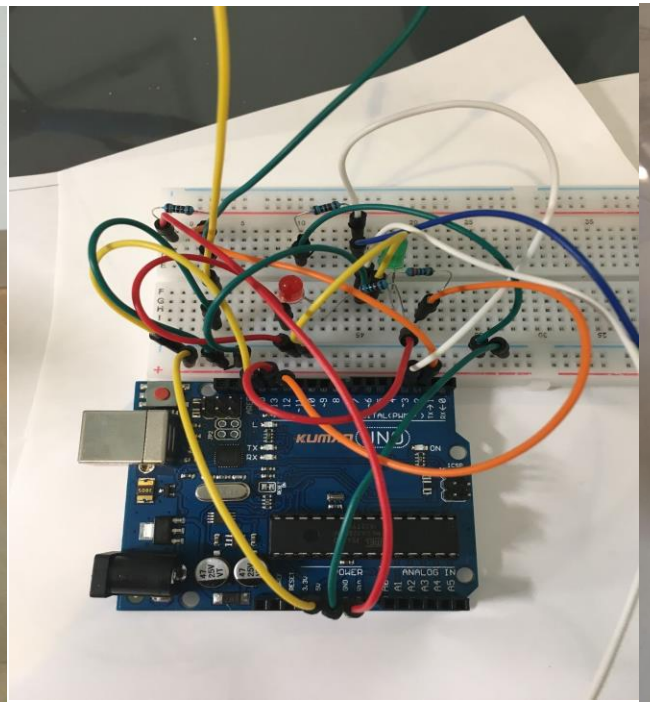


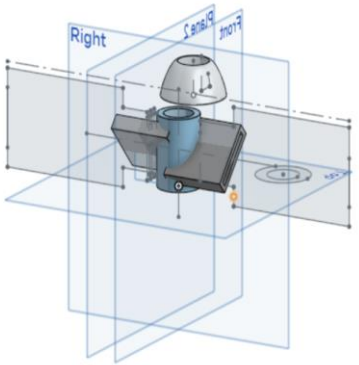


Early Rapid Prototyping

I used wood working, 3D printing, and Arduino in Early Prototypes





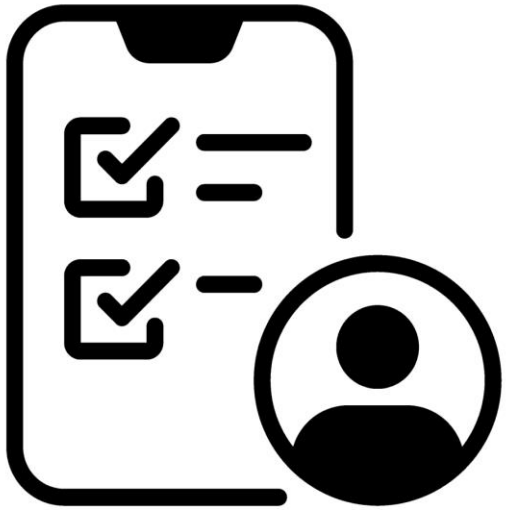


Initial User Testing & Feedback

Designed (CAD) & Built (Rapid Prototyping) a Full Scale Prototype

Methodology:

- Placed in different homes & rooms to simulate actual use conditions
- Conducted feedback and interaction sessions with potential consumers
- Observed:
  - Spatial fit within kitchens/rooms/apartments
  - User interaction and approach behavior
  - Initial perception and willingness to adopt



Findings:

- Large footprint created excessive visual and spatial dominance in the home
- Users were hesitant to adopt a large, unfamiliar appliance
- Perceived as a high-commitment product despite its functionality

Key Insight:

The large form factor created a significant barrier to adoption, leading me to push the design towards smaller, more adaptable, and flexible configurations that better fit within real home spaces.

Iteration: Moving to Smaller Form Factor

Designed (CAD), built, & tested an updated set of smaller porotypes

Approach:

- Identified opportunity to create a small-scale modular system that allows the product to adapt to the user
  - Uniform rectangular 6" x 6" x 14" profile
  - Enables users to start small and scale up based on space and needs

Initial Rapid Prototype Design:

- Developed multiple architectures and assemblies using **CAD** (Solidworks & NX)
- Iterated rapidly with early prototypes to test ergonomics, usability, and product-market fit
- Prototyped **electronic systems** using Arduino and raspberry Pi. Soldered lights and pump to relay terminals
- Developed geared water mixing system (designed custom gearbox) and a unique water flow geometry to maximize efficiency

User Testing:

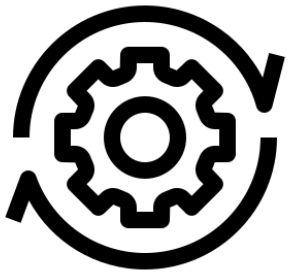
- Users responded positively to the simple, clean, and ergonomic form factor
- Confirmed smaller size works well for apartments and compact spaces

Final Design for Manufacturing & Scaling

Objective: Translate prototype into scalable, commercial-ready design

Manufacturing and Scaling Design

- Designed for Manufacturability, considering final manufacturing techniques:
  - **Large Scale Injection molding**
  - Vacuum Forming
  - CNC
  - Sheet Metal Forming
  - Stamping
  - Water Jet Cutting
- Used rapid prototyping techniques to quickly build initial prototypes
- Incorporated manufacturing processes (vacuum forming, injection molding design, CNC) to guide scalable commercial system development



Cost Optimization & Business Plan

- Redesigned to cut costs by 40% and assembly steps by 20%
- Determined estimated total production cost and created **marketing and sales plan** to ensure **profitability** and flexible profit margin
- Explored market opportunities, industry trends, and conducted **consumer studies** to validate adoption feasibility

Final GreenLoop Product

Explored varied colors and finishes to refine aesthetics and appeal to different consumer preferences

Internship Projects

- **NDA-safe:**
  - My apple work is highly restricted, All fan controller related materials in this presentation are conceptual to comply with NDA
  - My P&G work is also highly restricted as the products I designed haven't been released yet, so the related material is conceptual and I am not showing pictures or drawings

- **Apple:** The project I did with apple was a 6-month end-to-end new product development project going all the way from market/consumer to manufacturing and testing
- **P&G:** The project I did with P&G was a new product-development project intended to create new product feature, functionality, durability, and aesthetic design to branch Mr. Clean brand out into a new market segment I worked closely with consumrs rapidly iterating and optimizing

Next Gen Fan Controller

Project Goal:

- Develop the next-generation Fan Controller by defining key features and designing the device, system architecture, and assembly to realize the next-generation controller

Problem Definition

Identifying key user pain-points, market opportunities and selecting features/requirements

Design (CAD) Architectures

Design multiple architectures to enable features and compare

Iterate, Prototype, & Refine Design

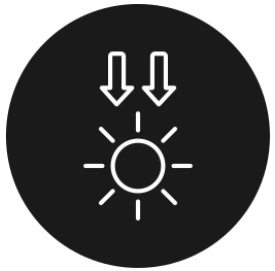
Quick design cycles, verify using Rapid Prototyping & performance/user testing

Manufacturing

Design for Manufacturing (Machining, & Injection molding), work to release PCB boards and compression molded parts

Next Gen Fan Controller

Consumer & Market Research Insights:



- Observed and interviewed users to better understand how they interact with the device to identify "pain points" and friction
- Users struggle to use controller in low-light settings
  - Device is frequently used in low light areas:
    - Just walked into a room, hasn't turned on light yet
    - In bed about to go to sleep
    - Watching a movie/tv
  - Can't tell which button to select because of low light

- Users complain about number of steps needed to replace Battery

- 60% of other premium controls on the market backlit their buttons

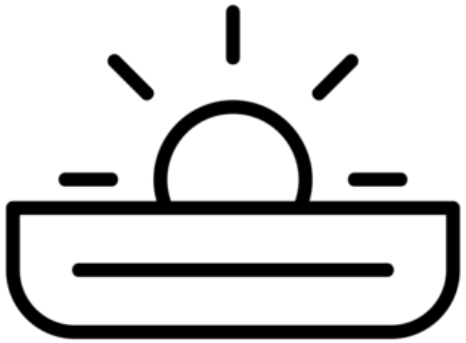
Self-Defined Project Requirements

Requirements & Constraints I Defined

- More Visible Buttons → Backlighting
- User facing simplified design adding new features to improve ergonomics
- Replaceable Battery (Must adhere to EU guidelines)
- Keep cost within 15% of current

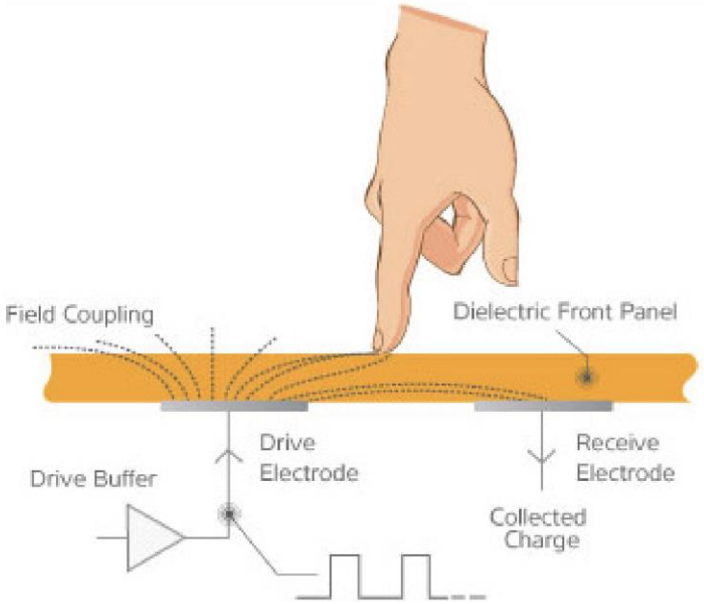
Next Gen Fan Controller

Chosen Features



Backlighting & New Button System

Novel Architecture and Functionality



Custom Capacitive Touch Sensing Capabilites

Removable Battery

Choosing Button Architecture

Backlit Button Exploration: Choosing Optimal Button Design

Final Choice

Snap Fit	Top Hat	Button Sheet
+	+ +	+ +
High Assembly Flexibility	Lower Light Leakage	Lower Cost
- -	- -	+ + +
High Part Count	High Part Count	Minimal Light Leakage
- - -		+ +
Severe Light Leakage		Lower Part Count
		+ +
		One-Step Assembly

Choosing Overall Device Architecture

Architecture Options

Final Choice

Clamshell	Back Panel	Unibody
+ Enables Button Sheet Assembly	+ + Simpler Manufacturing	+ + + Requires Snap-Fit
- - Too short screw bosses	+ + Enables Button Sheet Assembly	+ + No external seams
- - Large Tolerance Loop for Buttons	- - Ample space for fasteners	
	+ + Short Tolerance Loop for Buttons	

User Testing & Refinement:

Button Switch Weight

Goal: I wanted to ensure users feel a satisfying, high-quality "click" while preventing a "cheap" or mushy feel

- **What I did:**
  - Tested multiple button stiffnesses (metal domes) and observed how users interacted with different levels of resistance
- **The Result:**
  - I selected the user preferred mid-range "weight" that provided clear tactile feedback
- **The Impact:**
  - This balance ensured the remote felt premium to the touch while being stiff enough to prevent accidentally input if bumped in a bag or on a couch

Key Challenges and Specific Feature Explanations

Next Gen Fan Controller

Challenge: Capacitive Sensor Challenge

My design uses the entire anodized metal enclosure as a sensor, whenever the user touches the controller, it triggers backlighting

Challenge: Backlighting triggered only when full hand is wrapped around controller, User testing showed that most users hold control with only partial hand contact

Solution:

- Relocated MLB enclosure screw to reduce electromagnetic steel/battery interference with sensor and added insulating spacer above threading

Outcome:

- High sensitivity & consistent ergonomic touch sensing capability

Cost Reduction Efforts

Challenge: Backlighting adds too much manufacturing cost to maintain a marketable product price point

Challenge: Backlighting components (LEDs & sensors) add ~ $2 per unit

Solution:

- Redesigned battery removability (removable component) such that battery is directly secure to enclosure rather than to a battery enclosure
- Redesigned Back Panel for extrusion rather than full CNC (**cheaper manufacturing**)

Outcome:

- 3 injection molded parts removed (lower part count) ➔Lowered cost by $0.6
- Manufacturing method for enclosure adjusted ➔ Lowered cost by around 20%
- Estimated cost adjusted from + $2 to ~ + $0.2

Tolerance Analysis Work Example

Challenge: Button Gap Tolerance & Backlighting

Gap between button sheet flange & enclosure needs consistent gap spacing to allow for uniform backlighting

Challenge: Current Gap variation between buttons is too high (+/- 0.15 mm)

Solution:

- Clamped/Sandwiched MLB (contains dome switches) directly to top enclosure face using a plastic support piece

Outcome:

- +/- 0.06 mm gap variation
- Halved total screws required to secure MLB
- More rigid support behind buttons→ prevents dome-switch cracking

Next Gen Fan Controller Results

Results

1. Delivered a fully functional prototype that integrated novel backlighting and capacitive touch sensing into a **compact, consumer-ready form factor**
2. Cut total part count by 40% & sub assemblies by 60% directly **reducing cost and assembly complexity**, enabling new features while maintaining competitive market price
3. Designed and release 2 part compression molded buttons & worked with vendor to create custom manufactured MLBs

Learnings

1. Tolerance loops matter early
2. Low Fidelity prototypes help expose major issue early and quick
3. Early consideration of manufacturing constraints can shape design decisions, revealing trade-offs between architecture complexity, design, and cost efficiency

P&G Home Brands

New Product Development, Research, & Design Project

Note:
Unfortunately, I can't include sketches, CAD, pictures, or technical product specifics due to NDA

New Mr. Clean Brand Product

Designing a New Product Segment: From Market Opportunity & Consumer Needs to Functional Hardware

Role: Product Design & New Product Development Intern @ P&G

The Goal: Address market opportunity/gap not properly covered within the current brand lineup by developing a new product concept to capture it, using industrial design to shift consumer perception, mechanical design to improve performance, & leveraging intuitive cues to drive correct usage and functional trust

Strategy:

- Use consumer research to define a new set of "key user traits" and translate those into a distinct hardware architecture
- Use CAD, fluid modeling, and Performance testing to design and validate the product to maximize performance in new use space

Outcome:

- Delivered an "Irresistibly Superior" product that beat top competitors and other brand line products in terms of performance (2X) and consumer acceptance (4X)

Market Opportunity Analysis & Intial Concept Ideation

Identifying the White Space
Designed and conducted user research studies to map & understand current user behaviors and identify why existing products were failing to meet consumer needs within a specific sector.
Created Feature and decision plan to inform initial designs

Initial Rapid Prototyping:

- Came up with 15+ innovative designs and product concepts, mapped out key features and then rapidly iterated through hand-made prototypes
- **Goal:** These weren't just for "fit", they were tools to map out the strategic direction of the product's form and function before moving into high fidelity designs and detailed CAD

Market Opportunity Analysis & Intial Concept Ideation

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Product Design/Development Cycle

Starting with my first set of low fidelity prototypes, I looped through a rapid cycle of **prototyping**, **consumer testing**, **performance testing**, and form & **engineering design** update. Then finishing with a **manufacturing design** optimization cycle

Engineering Design

Developed product architecture and assemblies, creating detailed CAD models and validating mechanical performance through iterative design

Performance Testing

Design and develop performance testing methodology to validate prototype performance vs baseline and competitors

User Testing

Evaluate user reactions to different prototypes and evaluate how they use & interact with the product

Rapid Prototyping

Started with high quantity of handmade form & aesthetic focused low-fidelity prototypes, and then moved up to higher fidelity final prototypes

Integrating Performance, Aesthetics, and Architecture

To deliver a product that was "irresistible" (P&G likes this word) to the target consumer, I overhauled 4 key pillars of the design

Visual & Physical Language

Developed a **new form-factor** & aesthetic design that signaled to consumers its intent and performance in the new market space

Mechanical Performance

Engineered new design, architecture, manufacturing, and material choices that provided **superior functional results** in the targeted cleaning environment.

Ergonomic Integration

Refined the physical touchpoints to ensure the product felt **intuitive** and was used correctly during high-intensity use.

Accessory Recommendations

Suggested accessory product concepts to complement the core system and **strengthen its overall market positioning**

Validation and Strategic Impact

Results

1. Final "blind" user testing showed **4:1 preference** for new product
2. **20% improved performance** in final performance testing
3. Strategic Goal: My product aimed to shift the brand strategy from "maintenance" to **category expansion**, consumer & market tests validated this
4. Developed **product architecture and manufacturing workflow** to allow reuse of **80% of existing line equipment for new product**

Key Learnings

1. **Design as Permission**: Visual cues must "give permission" for new usage pattern, especially for risk adverse users
2. Identifying fatigue and failure points early is essential to build of off when **improving mechanical performance**
3. **Quick scrappy prototyping** validated the product DNA in hours to avoid weeks of CAD rework and ensure the later design is on track

Thank You!

Nice to see you again! Thank you for taking the time to read through my portfolio, I really appreciate it.

If you have any questions, ideas, or just want to talk, please feel free to reach out

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